SIDA（Slot-Internal Deepening Algorithm）完整框架藍圖（中文版）：附錄 F：Topological Template Classifier 實戰示例

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附錄 F：Topological Template Classifier 實戰示例

- 以「工業革命」為案例，示範如何匹配到「五階段需求擴張」拓撲模板（A1–A5）
- 展示不同文化下的相同拓撲映射（西方技術演進 ↔ 東方王朝盛衰循環）

好的，這是你要的附錄 F 草案。我把「Topological Template Classifier」的核心邏輯、A1–A5 模板定義、以及「工業革命」與「東方王朝循環」的對映都放進來，夠實戰、也方便你之後接資料跑模型。

1)「五階段需求擴張」拓撲模板（A1–A5）

拓撲語義：一個系統在「潛在需求 → 觸發 → 擴張 → 制度化/飽和 → 分歧/再啟動」的可重入拓撲。它不是時間表，而是相位拓撲；可在不同文明、產業上"嵌入對準"。

階段	拓撲名	場徵（訊號）	典型變量傾向
A1	潛勢聚積（Seeded Latency）	潛在需求高、供給受限、邊際創新零星	ΔI↑、S低、Jo零星
A2	觸發點（Ignition）	關鍵技術/制度突破，成本拐點出現	成本↓、專利/創新密度↑、採納初速↑
A3	基礎設施型擴張（Scale-out）	生產/運輸/金融基建密集投資，外部性放大	產出彈性↑、資本形成率↑、網絡度↑
A4	制度化與飽和（Institutionalization）	標準/法規/壟斷/工會成形，增長趨平	價差壓縮、集中度↑、波動率↓
A5	分歧與再啟動（Bifurcation/Reset）	新技術/邊陲市場分支、舊秩序僵化誘發重啟	新S曲線萌芽、舊曲線報酬遞減

變量記號可沿用你的框架：潛在差 **ΔI**，資源池 **S**，初級流通 **Jo** 等。

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2) Classifier：特徵擷取與判定規則（摘要）

輸入：跨時序語料 + 指標（價量、制度事件、專利/科舉、基建、衝突/罷工等）

核心特徵

- **ΔI proxy**：價格/工資剪刀差、城市糧價–工資比、能源單位成本。
- **Adoption kinks**：採納曲線（logistic）的一階/二階導數極值。

- **Infra index**：鐵路/運河/電網里程、煤鐵產量、保險與票據擴張。
- **Institutional density**：法規條目、工會/行會數、標準化事件密度。
- **Concentration & conflict**：產業集中度、壟斷案、罷工/騷亂頻次。
- **Narrative shift**：語料中框架詞轉向（「進步/效率」→「安全/規範」）。

簡化判定

- A1：ΔI 高、採納未現 S-curve、創新孤島；
- A2：成本/效率拐點 + 媒體/檔案出現「新機器/新法」的關鍵詞尖峰；
- A3：基建/金融中介同步加速（共振）且外部性詞頻上升（「聯通/速度」）；
- A4：成長斜率趨平 + 法規/工會 + 集中度上升 + 風險詞（「事故/壟斷」）；
- A5：新技術簇在邊陲崛起、舊核心增長降噪，敘事兩極化（「顛覆/守成」）。

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3) 案例一：工業革命 ↔ A1–A5

階段	西方技術演進（英國主軸為例）	代表訊號
A1 潛勢聚積	18 世紀前期：城市化上升、羊毛—棉布需求偏移、手工/行會供給受限	紡織品相對價格↑、行會規制詞頻↑、家庭手工占比高
A2 觸發點	飛梭、珍妮紡紗機、水力/馬力工場；瓦特改良蒸汽機（效率拐點）	能源單位成本↓、產出/工時比↑、專利簇集中在紡織/蒸汽
A3 基建擴張	礦—鐵—煤鏈條、機械廠、鐵路/運河網、合股公司與保險擴張	鐵/煤產量曲線陡升、里程數↑、資本形成率↑、票據流通↑
A4 制度化/飽和	工廠制度成型、童工/勞安法、反托拉斯雛形、標準化量具；企業集中	法規條目密度↑、產業 CR4↑、事故/工運詞頻↑、成長斜率放緩
A5 分歧/再啟動	電力化/化學工業（第二次工業革命）、美德後發追趕、殖民邊陲市場	新 S 曲線（電/化/內燃機）↑；舊鏈條報酬遞減、敘事分歧

> 若拉長到 20 世紀，可見 A5 觸發下一輪 A2：電氣化與大規模生產（福特主義）作為新拓撲的 ignition。

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4) 案例二：東方王朝盛衰循環 ↔ A1–A5

階段	東方王朝循環（概念化）	代表訊號
A1 潛勢聚積	末世亂局—治安/糧運/地權失序，秩序需求高	粮價波動↑、盜亂紀錄↑、地方檄文/饑饉詞頻↑
A2 觸發點	建朝／改制、均田/賦稅改革、軍政一體化（秩序效率拐點）	詔令密度↑、戶籍/地冊重編、徭役制度重構
A3 基建擴張	大運河/屯田/水利、科舉與官僚編制擴張、邊疆封貢體系整合	河工/倉儲紀錄↑、科舉錄取↑、人口與糧稅彈性↑
A4 制度化/飽和	官僚層肥厚、內卷化、冗員冗費、士紳地權集中、腐敗/禁令增多	赦令與禁令↑、田產集中↑、告訐/科場舞弊詞頻↑
A5 分歧/再啟動	天災疊加財政失衡、民變邊患、軍鎮化 → 舊序崩解，新政權在邊陲引爆	起義/兵變密度↑、邊鎮詞頻↑、稅糧斷供、政令失靈

同構對映（直觀）

- 技術—制度觸發（**A2**）：瓦特蒸汽機 vs. 均田/稅制重構 → 效率拐點。
- 基建—編制擴張（**A3**）：鐵路/保險/合股 vs. 運河/科舉/倉儲 → 傳輸/治理容量激增。
- 飽和—僵化（**A4**）：工廠規範/壟斷 vs. 官僚冗贅/地權集中 → 增長斜率趨平。
- 分歧—重啟（**A5**）：電氣化再啟 vs. 起義改朝 → 新 **S** 曲線的點火。

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5) **Disambiguation**：常見誤判與修正

- **A2 vs. A3**：若看到投資暴衝就判 A3 容易誤判；需同時見到系統性網絡外部性（運輸里程+金融中介+專業化產能）才算 A3。
- **A4 vs.** 需求周期低谷：短期景氣回落≠A4。A4 必須伴隨制度密度上升 **+** 集中度上升 **+** 成長斜率結構性變平。
- **A5 vs.** 僅技術分流：只有當「新曲線具自我維持的採納動力」且舊曲線出現報酬遞減，才標記 A5。

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6) **Classifier** 輸出（示例 **JSON**）


```
{
  "case": "Industrial Revolution (Britain core)",
  "window": "1700–1910",
  "topology_template": "Five-Phase Demand Expansion (A1–A5)",
  "phase_assignments": [
    {"phase": "A1", "span": "1700–1740", "confidence": 0.72, "signals": ["ΔI_high_textiles", "gui
    {"phase": "A2", "span": "1740–1775", "confidence": 0.81, "signals": ["steam_efficiency_kink",
    {"phase": "A3", "span": "1775–1850", "confidence": 0.88, "signals": ["rail_canal_mileage_up",
    {"phase": "A4", "span": "1850–1890", "confidence": 0.77, "signals": ["factory_laws_density_up
    {"phase": "A5", "span": "1890–1910", "confidence": 0.74, "signals": ["electrification_adoptic
  ],
  "notes": "A5 shows emergence of a new ignition sequence (Second IR), i.e., next-cycle A2."
}
```


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7) 你可以怎麼用

- 單一文明內多產業對準：同年代把紡織、鐵路、保險三條曲線對齊，觀察拓撲是否同拍（共振 → A3 強證）。
- 跨文明對位：把「運河—科舉—倉儲」指標標準化，與「鐵路—公司法—保險」對映，同時跑 A1–A5 打分。
- 敘事/法規語料做相位錨點：把「進步/效率/開放」與「安全/規範/禁令」語義比值視為相位指示燈，常可準確切 A2→A3→A4 的分界。

如果你要，我可以把這套判定規則抽成 **YAML/JSON** 規格（特徵 → 指標 → 條件 → 門檻），或補一張**Mermaid** 階段–訊號甘特圖，方便你直接接到 Slot/Template 雙層分類器。需要我加嗎？

太好了—我把 Slot Collapse Detector 的「特徵 → 指標 → 條件 → 門檻」抽成一份可機讀又可讀懂的規格。你可以直接拿去做參數檔（YAML/JSON 都行），也附了極簡 JSON Schema 方便驗

證。

YAML 規格（建議作為預設配置）

```yaml
version: 1.0
detector:
  name: SlotCollapseDetector
  description: 判定語義 slot 是否已收束（collapse）或陷入強吸引子（Strong Attractor）
  slots:
    default_window:
      min_turns: 6            # C1：資料量足夠
      max_window: 50          # 最長觀測窗
  features:
    # === 原始特徵（從語料/模型直接計算） ===
    - id: phase_alignment
      desc: slot 內各候選語義向量的相位/方向一致度（0–1）
      source: "cosine_mean_pairwise or circular-mean on θ"
    - id: slot_symmetry_index
      desc: 順/逆、肯/否、正/負等成對維度的平衡度（0 完全失衡；1 完全對稱）
      source: "paired-dimension balance over normalized counts"
    - id: entropy
      desc: 候選語義分佈熵（bits）
      source: "Shannon entropy over {intent/topic/template}"
    - id: entropy_delta
      desc: 近 k 回合熵的下降量（正值=收斂）
      source: "rolling diff of entropy"
    - id: turn_consistency
      desc: 連續回合輸出模板/拓撲的一致率（0–1）
      source: "template_id repeat ratio"
    - id: polarity_balance
      desc: 立場/情緒/價值極性的平衡指標（0–1）
      source: "sentiment/stance polarity distribution"
    - id: repair_rate
      desc: 自我修正/更正頻率（每 10 回合次數）
      source: "edit/correction markers"
    - id: novelty_rate
      desc: 新資訊/新證據引入比率（0–1）
      source: "unique facts/URLs/new claims / turn"
    - id: loop_score
      desc: 模板循環或語義回圈強度（0–1）
      source: "n-gram + template cycle detection"
    - id: response_latency_z
      desc: 反應延遲的標準分（越低=越穩定）
      source: "z-scored latency vs user/session baseline"
    - id: attractor_lexicon_hit
      desc: 強吸引子觸發詞/句式命中率（0–1）
      source: "keyword/pattern list match ratio"
    - id: contradiction_rate
      desc: 內在自相矛盾檢測率（0–1，越低越一致）
      source: "entailment/contradiction model"

  indicators:
    # === 由特徵綜合計分的「指標」 ===
    - id: symmetry_ok
      formula: "slot_symmetry_index >= θ.symmetry_min"
    - id: convergence_score
      formula: >
        0.35*phase_alignment
        + 0.25*min(1, max(0, entropy_delta / θ.entropy_delta_norm))
        + 0.20*turn_consistency
        + 0.10*(1 - contradiction_rate)
        + 0.10*(1 - response_latency_z_norm)
    - id: stability_score
      formula: >
```

```
      0.40*turn_consistency
      + 0.25*(1 - repair_rate_norm)
      + 0.20*(1 - loop_score)
      + 0.15*(1 - contradiction_rate)
  - id: novelty_guard
    formula: "novelty_rate >= θ.novelty_min"
  - id: strong_attractor_score
    formula: >
      0.45*loop_score
      + 0.25*attractor_lexicon_hit
      + 0.15*(1 - novelty_rate)
      + 0.15*turn_consistency

thresholds:
  # === 門檻（可按資料域微調） ===
  symmetry_min: 0.55
  entropy_delta_norm: 0.8        # 使 entropy_delta 正規化的標度
  convergence_pass: 0.68
  stability_pass: 0.70
  novelty_min: 0.18
  strong_attractor_warn: 0.62
  strong_attractor_fail: 0.75
  latency_z_clip: [-2.5, 2.5]

conditions:
  # === 判定步驟 C1–C4 ===
  - id: C1_data_sufficiency
    rule: "window_turns >= slots.default_window.min_turns"
    fail_state: "INSUFFICIENT_DATA"
  - id: C2_symmetry_gate
    rule: "symmetry_ok == true"
    on_fail:
      allow_override_if: "convergence_score >= thresholds.convergence_pass + 0.07"
      else_state: "ASYMMETRY_BLOCK"
  - id: C3_convergence_and_stability
    rule_all:
      - "convergence_score >= thresholds.convergence_pass"
      - "stability_score >= thresholds.stability_pass"
      - "novelty_guard == true"
    fail_state: "NOT_CONVERGED"
  - id: C4_strong_attractor_scan
    rule:
      - when: "strong_attractor_score >= thresholds.strong_attractor_fail"
        state: "STRONG_ATTRACTOR_LOCK"
      - when: "strong_attractor_score >= thresholds.strong_attractor_warn"
        state: "STRONG_ATTRACTOR_WARN"
      - else: "OK"

collapse_mapping:
  # === 輸出狀態映射 ===
  - when: "C1 && C2 && C3 && C4==OK"
    collapse_state: "COLLAPSED"
  - when: "C4==STRONG_ATTRACTOR_WARN"
    collapse_state: "COLLAPSED_BUT_FRAGILE"
  - when: "C4==STRONG_ATTRACTOR_LOCK"
    collapse_state: "ATTRACTOR_LOOP"
  - when: "C1 fail"
    collapse_state: "PENDING"
  - when: "C2 fail"
    collapse_state: "NEEDS_SYMMETRY"
  - when: "C3 fail"
    collapse_state: "NOT_CONVERGED"

normalization:
  # === 正規化建議 ===
  response_latency_z_norm:
```

```
    clip: "* thresholds.latency_z_clip"
    rescale: "map to [0,1] by percentile in-session"
  repair_rate_norm:
    base: "per 10 turns"
    rescale: "min-max in domain corpus"
```

對應 **JSON**（同一份配置）


```
{
  "version": "1.0",
  "detector": {
    "name": "SlotCollapseDetector",
    "description": "判定語義 slot 是否已收束（collapse）或陷入強吸引子（Strong Attractor）",
    "slots": {
      "default_window": { "min_turns": 6, "max_window": 50 }
    },
    "features": [
      { "id": "phase_alignment", "desc": "slot 內相位一致度（0–1）", "source": "cosine_mean_pairwi
      { "id": "slot_symmetry_index", "desc": "成對維度平衡度（0–1）", "source": "paired-dimension
      { "id": "entropy", "desc": "候選語義分佈熵（bits）", "source": "Shannon entropy" },
      { "id": "entropy_delta", "desc": "近 k 回合熵的下降量", "source": "rolling diff" },
      { "id": "turn_consistency", "desc": "模板一致率（0–1）", "source": "template repeat ratio" }
      { "id": "polarity_balance", "desc": "極性平衡（0–1）", "source": "stance/sentiment distribut
      { "id": "repair_rate", "desc": "自我修正頻率/10 turns", "source": "edit markers" },
      { "id": "novelty_rate", "desc": "新資訊比率（0–1）", "source": "unique facts/URLs" },
      { "id": "loop_score", "desc": "循環強度（0–1）", "source": "cycle detection" },
      { "id": "response_latency_z", "desc": "延遲 z 分數", "source": "z-score vs baseline" },
      { "id": "attractor_lexicon_hit", "desc": "吸引子詞命中率（0–1）", "source": "lexicon/pattern
      { "id": "contradiction_rate", "desc": "自相矛盾率（0–1）", "source": "NLI model" }
    ],
    "indicators": [
      { "id": "symmetry_ok", "formula": "slot_symmetry_index >= θ.symmetry_min" },
      {
        "id": "convergence_score",
        "formula": "0.35*phase_alignment + 0.25*min(1, max(0, entropy_delta / θ.entropy_delta_nor
      },
      {
        "id": "stability_score",
        "formula": "0.40*turn_consistency + 0.25*(1 - repair_rate_norm) + 0.20*(1 - loop_score) +
      },
      { "id": "novelty_guard", "formula": "novelty_rate >= θ.novelty_min" },
      {
        "id": "strong_attractor_score",
        "formula": "0.45*loop_score + 0.25*attractor_lexicon_hit + 0.15*(1 - novelty_rate) + 0.15
      }
    ],
    "thresholds": {
      "symmetry_min": 0.55,
      "entropy_delta_norm": 0.8,
      "convergence_pass": 0.68,
      "stability_pass": 0.70,
      "novelty_min": 0.18,
      "strong_attractor_warn": 0.62,
      "strong_attractor_fail": 0.75,
      "latency_z_clip": [-2.5, 2.5]
    },
    "conditions": [
      { "id": "C1_data_sufficiency", "rule": "window_turns >= slots.default_window.min_turns", "f
      {
        "id": "C2_symmetry_gate",
        "rule": "symmetry_ok == true",
        "on_fail": { "allow_override_if": "convergence_score >= thresholds.convergence_pass + 0.0
      },
      {
```

```
        "id": "C3_convergence_and_stability",
        "rule_all": [
          "convergence_score >= thresholds.convergence_pass",
          "stability_score >= thresholds.stability_pass",
          "novelty_guard == true"
        ],
        "fail_state": "NOT_CONVERGED"
      },
      {
        "id": "C4_strong_attractor_scan",
        "rule": [
          { "when": "strong_attractor_score >= thresholds.strong_attractor_fail", "state": "STRON
          { "when": "strong_attractor_score >= thresholds.strong_attractor_warn", "state": "STRON
          { "else": "OK" }
        ]
      }
    ],
    "collapse_mapping": [
      { "when": "C1 && C2 && C3 && C4==OK", "collapse_state": "COLLAPSED" },
      { "when": "C4==STRONG_ATTRACTOR_WARN", "collapse_state": "COLLAPSED_BUT_FRAGILE" },
      { "when": "C4==STRONG_ATTRACTOR_LOCK", "collapse_state": "ATTRACTOR_LOOP" },
      { "when": "C1 fail", "collapse_state": "PENDING" },
      { "when": "C2 fail", "collapse_state": "NEEDS_SYMMETRY" },
      { "when": "C3 fail", "collapse_state": "NOT_CONVERGED" }
    ],
    "normalization": {
      "response_latency_z_norm": { "clip": "* thresholds.latency_z_clip", "rescale": "map to [0,1
      "repair_rate_norm": { "base": "per 10 turns", "rescale": "min-max in domain corpus" }
    }
  }
}
```

迷你 **JSON Schema**（驗證配置檔用）

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "SlotCollapseDetectorConfig",
  "type": "object",
  "required": ["version", "detector"],
  "properties": {
    "version": { "type": "string" },
    "detector": {
      "type": "object",
      "required": ["name", "features", "indicators", "thresholds", "conditions", "collapse_mappin
      "properties": {
        "name": { "type": "string" },
        "slots": { "type": "object" },
        "features": {
          "type": "array",
          "items": { "type": "object", "required": ["id"], "properties": {
            "id": { "type": "string" },
            "desc": { "type": "string" },
            "source": { "type": "string" }
          } }
        },
        "indicators": {
          "type": "array",
          "items": { "type": "object", "required": ["id","formula"], "properties": {
            "id": { "type": "string" },
            "formula": { "type": "string" }
          } }
        },
        "thresholds": { "type": "object", "additionalProperties": { "type": ["number","array"] }
        "conditions": {
```

```
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          "items": { "type": "object", "required": ["id"], "properties": {
            "id": { "type": "string" },
            "rule": { "type": "string" },
            "rule_all": { "type": "array", "items": { "type": "string" } },
            "fail_state": { "type": "string" },
            "on_fail": { "type": "object" },
            "rule_any": { "type": "array", "items": { "type": "string" } }
          } }
        },
        "collapse_mapping": {
          "type": "array",
          "items": { "type": "object", "required": ["when","collapse_state"], "properties": {
            "when": { "type": "string" },
            "collapse_state": { "type": "string" }
          } }
        },
        "normalization": { "type": "object" }
      }
    }
  }
}
```

使用小提醒

- 想更嚴格：把 `symmetry_min` 拉高到 0.6–0.65；或把 `convergence_pass` 調到 0.72。
- 想更敏捷：放寬 `min_turns` 到 4，並降低 `stability_pass` 到 0.65。
- 強吸引子偵測敏感度：提高或降低 `strong_attractor_warn/fail` 即可。

要不要我幫你把這份規格塞進你的現有程式（或做個最小可跑的檢測器）？

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This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

I am merely a midwife of knowledge.